

Containerization & DevOps Lab

Document: Lab6.md | Containerization & DevOps Coursework

Experiment 6A

Title: Comparison of Docker Run and Docker Compose

PART A - THEORY

1. Objective

To understand the relationship between `docker run` and Docker Compose, and to compare their configuration syntax and use cases.

2. Background Theory

2.1 Docker Run (Imperative Approach)

The `docker run` command is used to create and start a container from an image. It requires explicit flags for:

- * Port mapping (`-p`)
- * Volume mounting (`-v`)
- * Environment variables (`-e`)
- * Network configuration (`--network`)
- * Restart policies (`--restart`)
- * Resource limits (`--memory` , `--cpus`)
- * Container name (`--name`)

This approach is imperative, meaning you specify step-by-step instructions.

Example:

```
docker run -d \  
  --name my-nginx \  
  -p 8080:80 \  
  -v ./html:/usr/share/nginx/html \  
  -e NGINX_HOST=localhost \  
  --restart unless-stopped \  
  nginx:alpine
```

2.2 Docker Compose (Declarative Approach)

Docker Compose uses a YAML file (`docker-compose.yml`) to define services, networks, and volumes in a structured format. Instead of multiple `docker run` commands, a single command is used: `docker compose up -d` . Compose is declarative, meaning you define the desired state of the application.

Equivalent Compose file:

```
version: '3.8'
services:
  nginx:
    image: nginx:alpine
    container_name: my-nginx
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html
    environment:
      NGINX_HOST: localhost
    restart: unless-stopped
```

3. Mapping: Docker Run vs Docker Compose

- * `-p 8080:80` equivalent to `ports:`
- * `-v host:container` equivalent to `volumes:`
- * `-e KEY=value` equivalent to `environment:`
- * `--name` equivalent to `container_name:`
- * `--network` equivalent to `networks:`
- * `--restart` equivalent to `restart:`
- * `--memory` equivalent to `deploy.resources.limits.memory`
- * `--cpus` equivalent to `deploy.resources.limits.cpus`

4. Advantages of Docker Compose

1. Simplifies multi-container applications
2. Provides reproducibility
3. Version controllable configuration
4. Unified lifecycle management
5. Supports service scaling (Example: `docker compose up --scale web=3`)

PART B - PRACTICAL TASK

Task 1: Single Container Comparison

Step 1: Run Nginx Using Docker Run

Execute:

```
yuvraj_singh@LUCIF3R 'exp6$ docker run ~d \
--name lab-nginx \
-p 8081:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx:alpine
4524adc22ad136ff5292e6d25cf1f713d4d64a1b59d47e0c7f5aa69ca35042c4
```

Verify: `docker ps`

```
yuvraj_singh@LUCIF3R:~ exp6$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
4524adc22ad1   nginx:alpine  "/docker-entrypoint..." About a minute ago Up About a minute  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp lab-nginx
```

Access: `http://localhost:8081`



Stop and remove container:

```
yuvraj_singh@LUCIF3R 'exp6$ docker stop lab-nginx
lab-nginx
yuvraj_singh@LUCIF3R:~/exp6$ docker rm lab-nginx
lab-nginx
```

Step 2: Run Same Setup Using Docker Compose

Create `docker-compose.yml` :

```
GNU nano 7.2
version: '3.8'

services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```

Run: `docker compose up -d`

```
yuvraj_singh@LUCIF3R exp6$ docker compose up -d
WARN[0000] Ihome yuvraj_singh/exp6/docker-compose yml: the attribute version is obsolete it will be ignored please remove it to avoid potential confusion
[+] up 1/1
✓Container lab-nginx Started 0.3s
yuvraj_singh@LUCIF3R $ docker compose ps
WARN[0000] Ihome yuvraj_singh/exp6/docker-compose yml: the attribute version ' is obsolete , it will be ignored, please remove it to avoid potential confusion
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
lab-nginx nginx:alpine "/docker-entrypoint..." nginx 22 seconds ago Up 21 seconds 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
yuvraj_singh@LUCIF3R exp6$ docker compose down
WARN[0000] Ihome yuvraj_singh/ 'exp6/docker-compose yml the attribute 'version is obsolete it will be ignored please remove it to avoid potential confusion
[+] down 2/2
✓Container lab-nginx Removed 0.6s
✓Network exp6_default Removed 0.3s
```

Verify: `docker compose ps`

Stop: `docker compose down`

Task 2: Multi-Container Application

Objective: Deploy WordPress with MySQL using:

1. Docker Run (manual way)
2. Docker Compose (structured way)

A. Using Docker Run

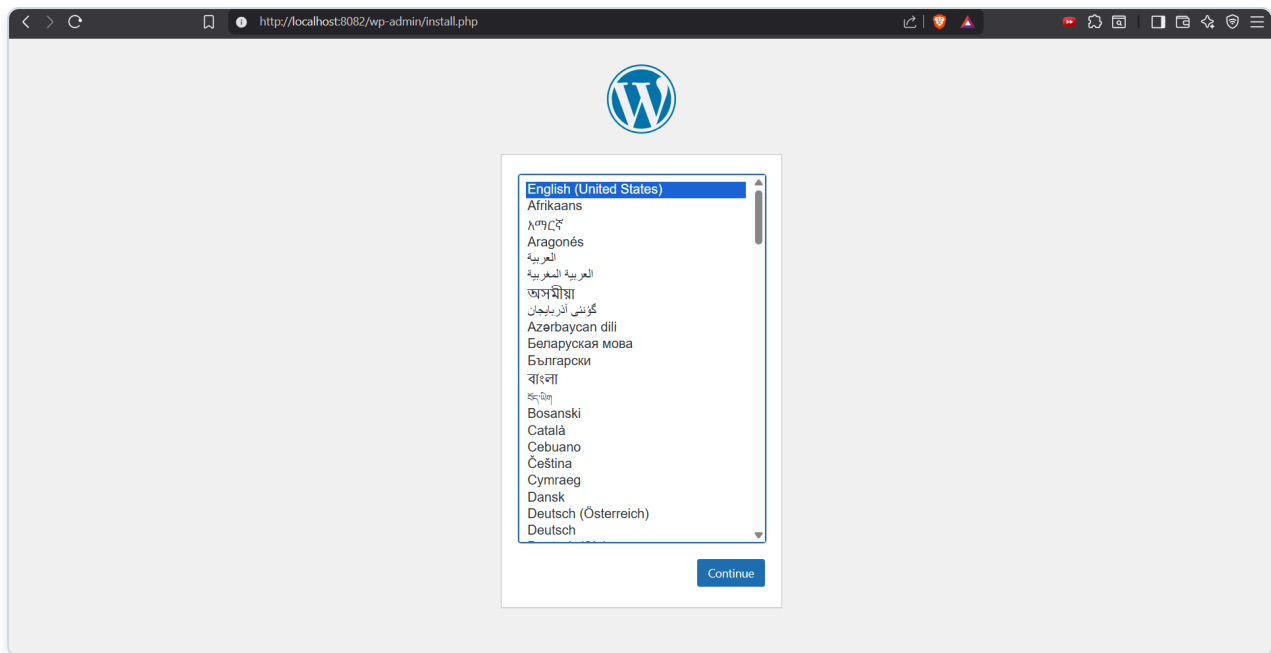
1. Create network: `docker network create wp-net`
2. Run MySQL:

```
yuvraj_singh@LUCIF3R $ docker run -d \
--name mysql \
--network wp-net \
-e MYSQL_ROOT_PASSWORD=secret \
-e MYSQL_DATABASE=wordpress \
mysql:5.7
c4b381ffdb7b8d8b14832d7bfde88e28c1f10e2b63531464f30566f7fb1e1ab8
```

3. Run WordPress:

```
yuvraj_singh@LUCIF3R exp6$ docker run -d \
--name wordpress \
--network wp-net \
-p 8082:80 \
-e WORDPRESS_DB_HOST=mysql \
-e WORDPRESS_DB_PASSWORD=secret \
wordpress:latest
477bc7b9a25c4a9746e3fdc212e36a720d62916c863d501b835001317ec7b0fa
```

Test: `http://localhost:8082`



B. Using Docker Compose

Create `docker-compose.yml` :

```
GNU nano 7.2
services:
  webapp:
    image: node.18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: "false"
    restart: unless-stopped
```

Run: `docker compose up -d`

```
yuvraj singh@LUCIF3R exp6s docker compose up -d
WARN[0000] Ihome yuvraj_singh/exp6/docker-compose.yml: the attribute version is obsolete it will be ignored , please remove it to avoid potential confusion
[*] up 2/2
✓Container exp6-mysql-1 Running 0.0s
✓Container exp6-wordpress-1 Started 0.4s
```

Stop: `docker compose down -v`

```
yuvraj singh@LUCIF3R exp6s docker compose down -v
WARN[0000] Ihome yuvraj_singh/exp6/docker-compose.yml: the attribute version is obsolete it will be ignored please remove it to avoid potential confusion
[*] down 4/4
✓Container exp6-wordpress-1 Removed 1.4s
✓Container exp6-mysql-1 Removed 1.5s
✓Volume exp6_mysql_data Removed 0.1s
✓Network exp6_default Removed 0.3s
```

PART C - CONVERSION & BUILD-BASED TASKS

Task 3: Convert Docker Run to Docker Compose

Problem 1: Basic Web Application

Given Docker Run Command:

```
docker run -d \
  --name webapp \
  -p 5000:5000 \
  -e APP_ENV=production \
  -e DEBUG=false \
  --restart unless-stopped \
  node:18-alpine
```

Student Task:

1. Write an equivalent docker-compose.yml

```
GNU nano 7.2
services:
  webapp:
    image: node.18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: "false"
    restart: unless-stopped
```

2. Ensure: Same container name, same port mapping, same environment variables, same restart policy.
3. Run using: `docker compose up -d`

```
yuvraj_singh@LUCIF3R:~/exp6$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED
STATUS              PORTS
webapp              node:18-alpine       "docker-entrypoint.s..." webapp     About a minute ago
Restarting (0) 38 seconds ago
```

4. Verify using: `docker compose ps`

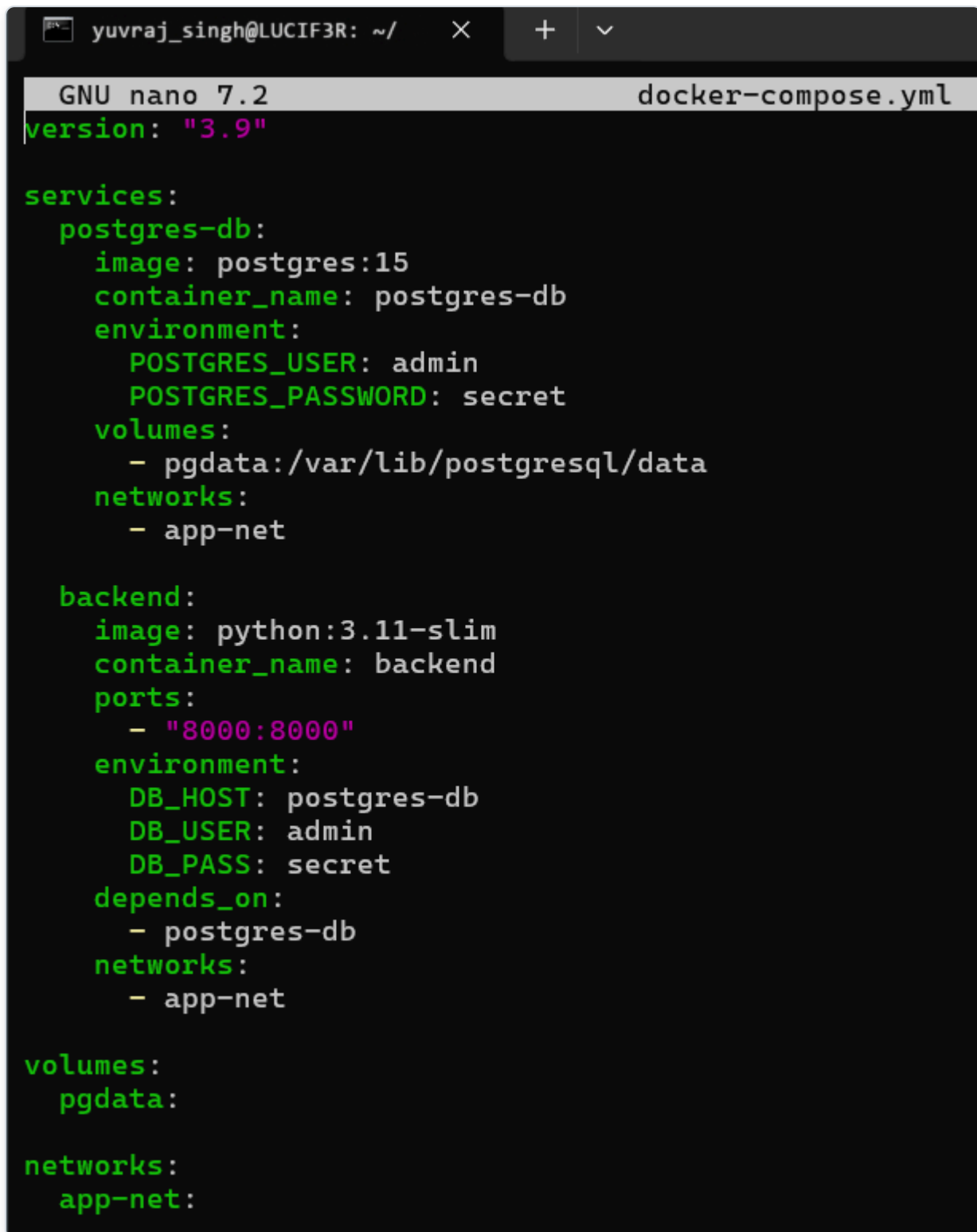
Problem 2: Volume + Network Configuration

Given Docker Run Commands:

```
docker network create app-net
docker run -d \
  --name postgres-db \
  --network app-net \
  -e POSTGRES_USER=admin \
  -e POSTGRES_PASSWORD=secret \
  -v pgdata:/var/lib/postgresql/data \
  postgres:15
docker run -d \
  --name backend \
  --network app-net \
  -p 8000:8000 \
  -e DB_HOST=postgres-db \
  -e DB_USER=admin \
  -e DB_PASS=secret \
  python:3.11-slim
```

Student Task:

1. Create a single docker-compose.yml file that defines both services, creates named volume pgdata, creates custom network app-net, and uses depends_on.

A screenshot of a terminal window with a dark background. The window title bar shows 'yuvraj_singh@LUCIF3R: ~/'. The terminal content shows the 'nano' editor editing 'docker-compose.yml'. The file content is a Docker Compose configuration for two services: 'postgres-db' and 'backend'. The 'postgres-db' service uses the 'postgres:15' image, sets environment variables for 'POSTGRES_USER' (admin) and 'POSTGRES_PASSWORD' (secret), and mounts a volume 'pgdata' to '/var/lib/postgresql/data'. The 'backend' service uses the 'python:3.11-slim' image, sets environment variables for 'DB_HOST', 'DB_USER', and 'DB_PASS', depends on the 'postgres-db' service, and also connects to the 'app-net' network. Both services are connected to the 'app-net' network. At the bottom, the 'volumes' and 'networks' sections are partially visible.

```
GNU nano 7.2 docker-compose.yml
version: "3.9"

services:
  postgres-db:
    image: postgres:15
    container_name: postgres-db
    environment:
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: secret
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks:
      - app-net

  backend:
    image: python:3.11-slim
    container_name: backend
    ports:
      - "8000:8000"
    environment:
      DB_HOST: postgres-db
      DB_USER: admin
      DB_PASS: secret
    depends_on:
      - postgres-db
    networks:
      - app-net

volumes:
  pgdata:

networks:
  app-net:
```

2. Bring up services using one command.

```

yuvraj singh@LUCIF3R = 'exp6s docker compose up ~d
WARN[0000] /home/yuvraj_singh/exp6/docker-compose yml the attribute version is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 6/7
  ✓ Image python:3.11-slim Pulled 8.4s
  ✓ Network exp6_app-net Creating 0.0s
[+] up 10/10 found orphan containers ([webapp]) for this project. If you removed or r
  ✓ Image python:3.11-slim Pulled 8.4s
  ✓ Network exp6_app-net Created 0.0s
  ✓ Volume exp6_pgdata Created 0.0s
  ✓ Container postgres-db Started 0.3s
  ✓ Container backend Started 0.3s
yuvraj_singh@LUCIF3R:~/exp6s docker compose down FV ~-remove-orphans
WARN[0000] /home/yuvraj_singh/exp6/docker-compose yml the attribute version is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 5/5
  ✓ Container backend Removed 0.0s
  ✓ Container postgres-db Removed 0.2s
  ✓ Container webapp Removed 0.0s
  ✓ Volume exp6_pgdata Removed 0.0s
  ✓ Network exp6_app-net Removed 0.3s

```

3. Stop and remove everything properly.

Task 4: Resource Limits Conversion

Given Docker Run Command:

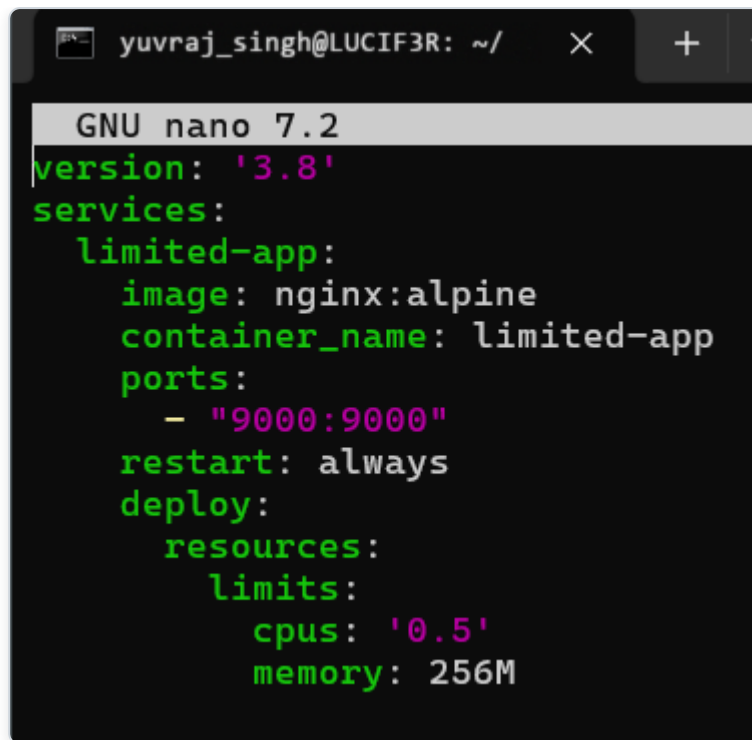
```

docker run -d \
  --name limited-app \
  -p 9000:9000 \
  --memory="256m" \
  --cpus="0.5" \
  --restart always \
  nginx:alpine

```

Student Task:

1. Convert this to Docker Compose.
2. Add resource limits using: `deploy > resources > limits`.



```
yuvraj_singh@LUCIF3R: ~/
GNU nano 7.2
version: '3.8'
services:
  limited-app:
    image: nginx:alpine
    container_name: limited-app
    ports:
      - "9000:9000"
    restart: always
    deploy:
      resources:
        limits:
          cpus: '0.5'
          memory: 256M
```

PART D - USING DOCKERFILE INSTEAD OF STANDARD IMAGE

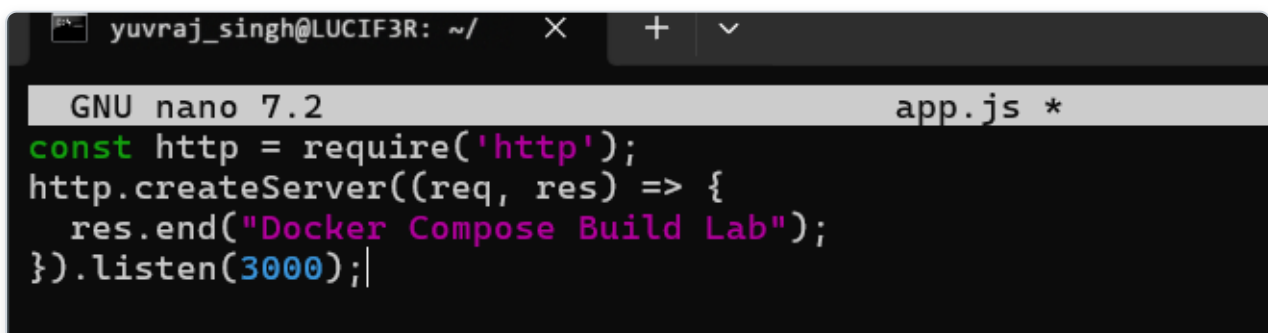
Task 5: Replace Standard Image with Dockerfile (Node App)

Instead of using prebuilt images like `image: node:18-alpine`, students must create their own Dockerfile, build image using Compose, and run container from that build.

Scenario: You are given `docker run -d -p 3000:3000 node:18-alpine`. Now you must:

1. Create a simple Node.js app
2. Write a Dockerfile
3. Use Docker Compose build: option

Step 1: Create app.js



```
yuvraj_singh@LUCIF3R: ~/
GNU nano 7.2 app.js *
const http = require('http');
http.createServer((req, res) => {
  res.end("Docker Compose Build Lab");
}).listen(3000);
```

Step 2: Create Dockerfile

```

GNU nano 7.2 Dockerfile *
FROM node:18-alpine
WORKDIR /app
COPY app.js
EXPOSE 3000
CMD ["node", "app.js"]

```

Step 3: Create docker-compose.yml

```

yuvraj_singh@LUCIF3R: ~/
GNU nano 7.2 docker-compose.yml *
version: '3.8'
services:
  nodeapp:
    build:
      context:
        dockerfile: Dockerfile
    container_name: custom-node-app
    ports:
      - "3000:3000"

```

Student Task:

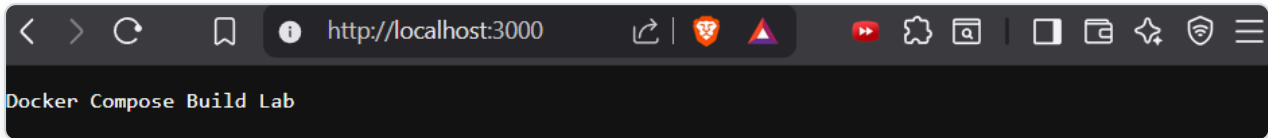
1. Build and run using: `docker compose up --build -d`

```

yuvraj_singh@LUCIF3R: exp6$ docker compose up --build -d
WARN[0000] Ihome/yuvraj_singh exp6/docker-compose yml: the attribute version 1s
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 0.3s (10/10) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 500B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 119B 0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc 0.0s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 153B 0.0s
=> [2/3] WORKDIR /app 0.0s
=> [3/3] COPY app.js . 0.0s
=> exporting to image 0.1s
=> => exporting layers 0.0s
=> => exporting manifest sha256:93f75belad4c7ffa0fc726289f693f58118c64ab062 0.0s
=> => exporting config sha256:12ab4374da5d73131af5014cb04a563a3052e2edc22e9 0.0s
=> => exporting attestation manifest sha256:c69b7c5d0ea664ee5204378ac787822 0.0s
=> => exporting manifest list sha256:7fd0f369196c32d6ff7be7317c88be723d7ca 0.0s
=> => naming to docker.io/library/exp6-nodeapp:latest 0.0s
=> => unpacking to docker.io/library/exp6-nodeapp:latest 0.0s
=> resolving provenance for metadata file 0.0s
[+] up 2/2
✔Image exp6-nodeapp Built 0.4s
✔Container custom-node-app Started 0.2s

```

2. Verify in browser: `http://localhost:3000`



3. Modify message in app.js, rebuild and observe changes.

4. Explain difference between `image:` and `build:`

Ans: `image:` is the specification of which image to use, whereas `build:` is used to tell which `Dockerfile` to use to build the image that will be used to run the container.

Task 6: Multi-Stage Dockerfile with Compose

Requirement: Create a simple Python FastAPI or Node production-ready app using:

- * Multi-stage Dockerfile
- * Smaller final image
- * Use Compose to build it

Must:

1. Write multi-stage Dockerfile

```
yuvraj_singh@LUCIF3R: ~/   
GNU nano 7.2 Dockerfile   
FROM node:18-alpine AS builder   
WORKDIR /app   
COPY package.json ./   
RUN npm install   
COPY . .   
   
FROM node:18-alpine   
WORKDIR /app   
COPY --from=builder /app .   
EXPOSE 8080   
CMD ["npm", "start"]
```

2. Use build: in Compose

```
GNU nano 7.2                                docker-compose.yml *
version: '3.8'
services:
  app:
    build:
      context: .
    ports:
      - "8080:8080"
    environment:
      - NODE_ENV=production
    volumes:
      - .:/app|
```

3. Add environment variables
4. Add volume mount for development mode
5. Compare image size: `docker images`

Experiment 6 B

Multi-Container Application using Docker Compose (WordPress + Database)

1. Objective

To deploy a multi-container application using Docker Compose, consisting of:

- * WordPress (frontend + PHP)
- * MySQL database (backend)

Also: Understand container networking & volumes, learn how to scale services, compare with Docker Swarm for production deployment.

2. Prerequisites

- * Docker installed
- * Docker Compose (comes with modern Docker)
- * Basic understanding of containers

3. Architecture Overview

User (Browser) -> WordPress Container -> MySQL Container -> Persistent Volume (Database Storage)

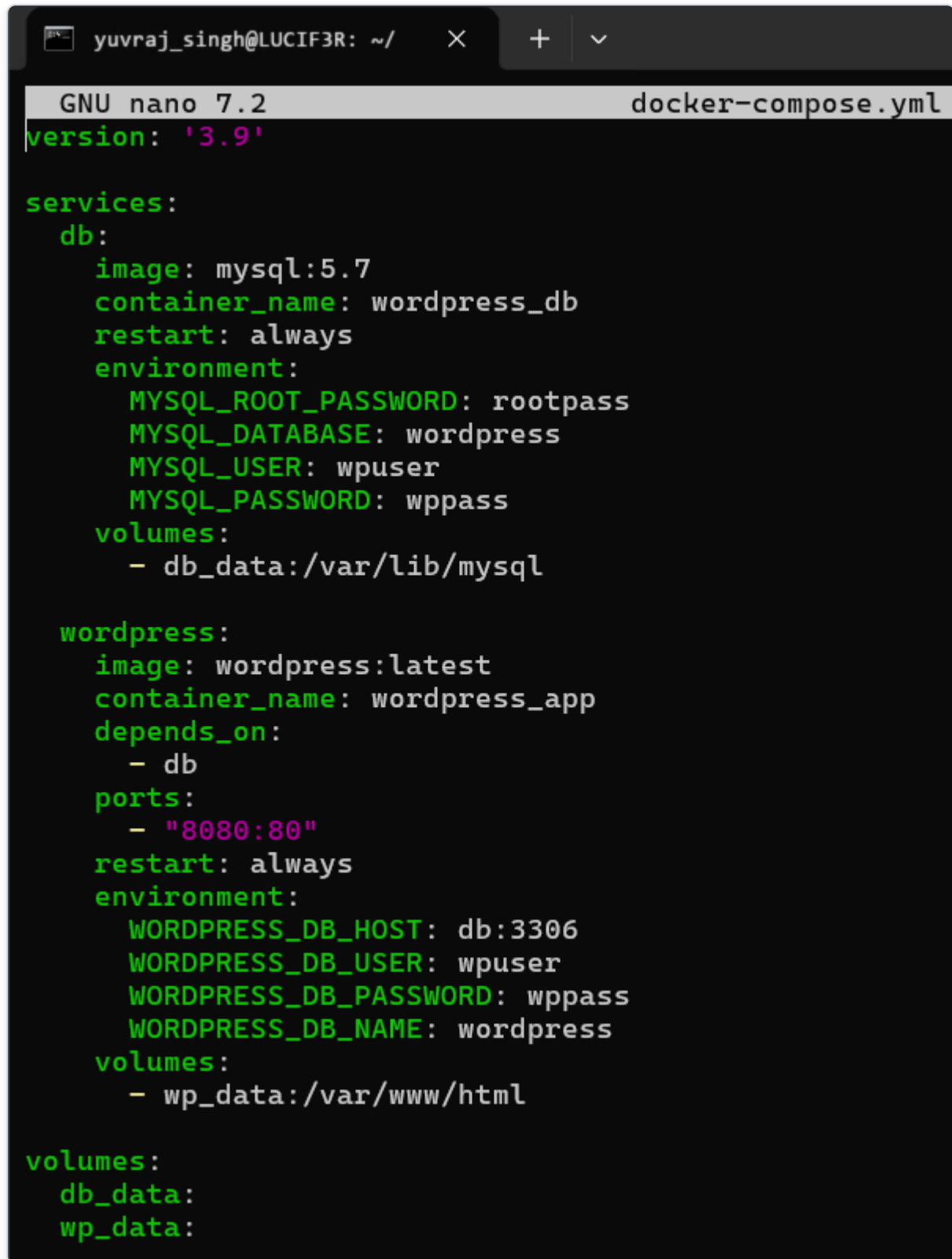
- * WordPress connects to MySQL using service name (DNS inside Docker network)
- * Data is persisted using volumes

Steps

Step 1: Create Project Directory

```
mkdir wp-compose-lab
cd wp-compose-lab
```

Step 2: Create docker-compose.yml



The screenshot shows a terminal window with a dark background. The title bar indicates the user is yuvraj_singh@LUCIF3R in the home directory. The terminal is running GNU nano 7.2, editing a file named docker-compose.yml. The content of the file is a Docker Compose configuration for a WordPress application and its database.

```
GNU nano 7.2 docker-compose.yml
version: '3.9'

services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

Explanation of Key Sections

- * **services:** Defines containers (db -> MySQL database, wordpress -> application)
- * **depends_on:** Ensures DB starts before WordPress
- * **environment:** Used to configure DB credentials and connection
- * **volumes:** Persist data even if containers are deleted
- * **ports:** Exposes WordPress on `http://localhost:8080`

Step 3: Start Application

```
docker-compose up -d
```

```
yuvraj_singh@LUCIF3R: exp6 wp compose-labf docker compose up ~d
WARN[0000] Ihome/yuvraj_singh/exp6/wp-compose-Lab/docker-compose .yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 5/5
✓Network wp-compose-lab_default Created 0.0s
✓Volume wp-compose-lab_db_data Created 0.0s
✓Volume wp-compose-lab_wp_data Created 0.0s
✓Container wordpress_db Started 0.2s
✓Container wordpress_app Started 0.3s
```

What happens: Images are pulled, Network is created, Containers are started, DNS-based service discovery enabled.

Step 4: Verify Containers

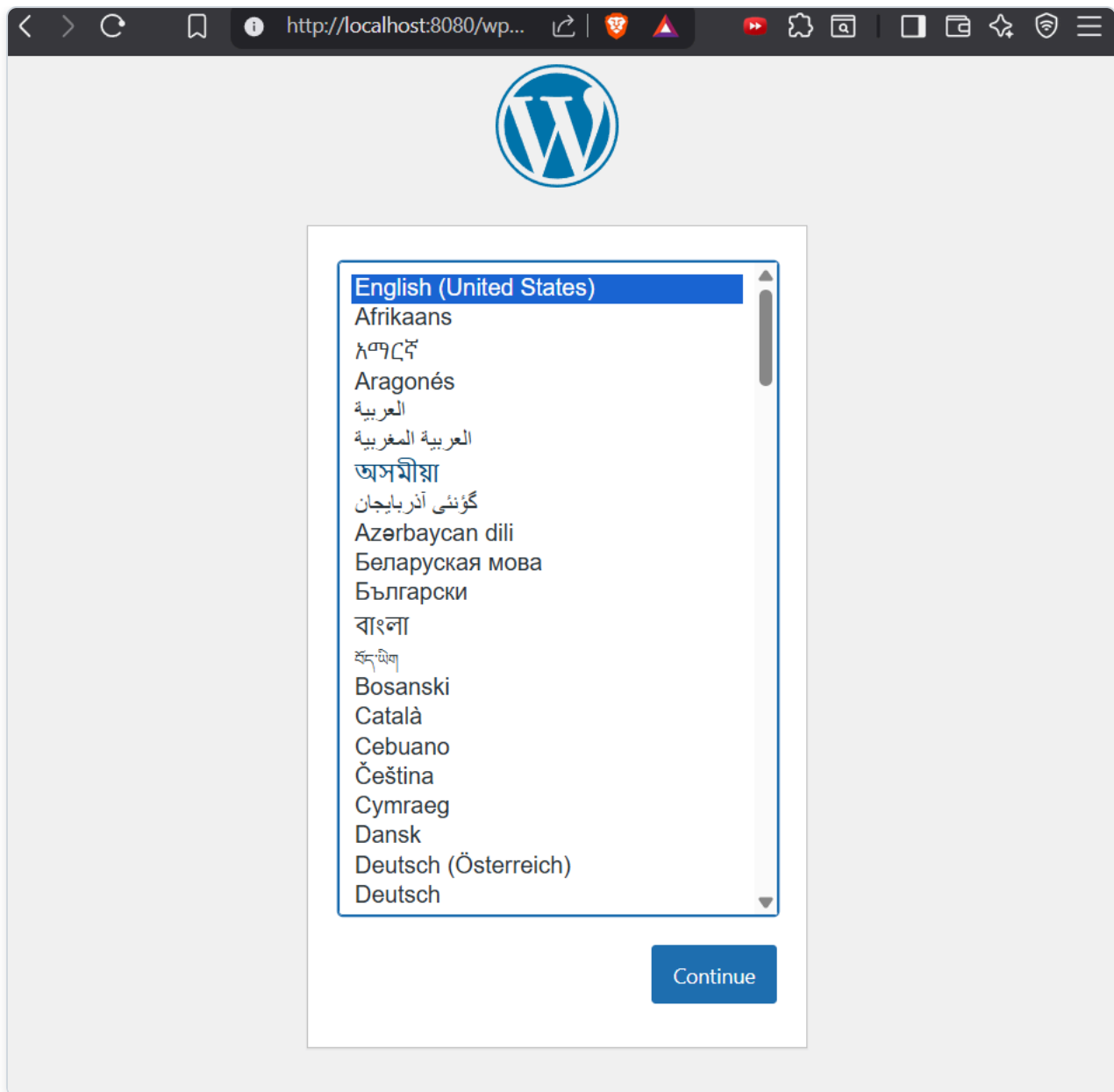
```
docker ps (Expected: wordpress_app, wordpress_db)
```

```
yuvraj_singh@LUCIF3R: exp6 wp compose-labf docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
342407ab9421	wordpress:latest	"docker-entrypoint.s..."	7 seconds ago	Up 7 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	wordpress_app
a17a29199d4d	mysql:5.7	"docker-entrypoint.s..."	7 seconds ago	Up 7 seconds	3306/tcp, 33060/tcp	wordpress_db
36c59d832bdb	exp6-nodeapp	"docker-entrypoint.s..."	17 minutes ago	Up 17 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	custom-node-app
3e201313ac91	wordpress:latest	"docker-entrypoint.s..."	About an hour ago	Up About an hour	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp	wordpress
55e15ecaf9f0	mysql:5.7	"docker-entrypoint.s..."	About an hour ago	Up About an hour	3306/tcp, 33060/tcp	mysql

Step 5: Access WordPress

Open browser: `http://localhost:8080` (Complete setup, enter title, user, password)



Step 6: Check Volumes

`docker volume ls` (db_data -> database persistence, wp_data -> WordPress files)

```
yuvraj_singh@LUCIF3R:~/exp6 Wp-compose- ab$ docker vo Lume IS
DRIVER      VOLUME NAME
local       5b25b03fcbd30f072a3c2841dda282dbc03630272a856fb95892789634a95915
local       8e22d80ef6accdb472c02d4a8b7c977ec2d374b0a59c165d5ea0815832f25275
local       9a116c9437d1d0b1ec608b91978e21ae2d64cc0a1c8c94c2a993e2dbdbcd89da
local       48fdefa035bc8abd516b974b005e8c2da1eec7fd0b9372555a764f8bacfd27d7
local       441ed787faa660c56d13c08ea1becaa8729a414d1c409eff372dd7baf18d6e62
local       app-volume
local       cb922b95c451c4a32b282b95e1eb8b5efb6337e49bb9cfc333abc9f431262cf7
local       e1f81d6fc8ca4a1e484eb6bbe4064cc5320e9fe4d3a1c0d865628a808ec3cdad
local       eeee3012b86a1237a291321574d2a1bf1cafb9eb9f44f552ec6bb63c2d946b9
local       fceffbc8008555e00094209b8fcd72118ea38a7d94f8b59aff82f4fc4bde0f57
local       mydata
local       mysql-data
local       pgdata
local       redis-data
local       wp-compose-lab_db_data
local       wp-compose-lab_wp_data
```

Step 7: Stop Application

`docker-compose down` (Containers removed, Volumes remain intact)

```
vuvrai singh@LUCIF3R: exp6/wp-compose-lab$ docker compose down
WARN[0000] /home/yuvraj_singh/exp6/wp-compose-Lab/docker-compose yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 3/3
✓Container wordpress_app      Removed      1.3s
✓Container wordpress_db      Removed      1.7s
✓Network wp-compose-lab_default Removed      0.2s
```

5. Scaling in Docker Compose

Method 1: Scale WordPress Containers

`docker-compose up --scale wordpress=3`

```
vuvrai singh@LUCIF3R: exp6/wp-compose-lab$ docker compose up --scale wordpress=3
WARN[0000] /home/yuvraj_singh/exp6/wp-compose-Lab/docker-compose yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 2/2
✓Network wp-compose-lab_default Created      0.0s
✓Container wordpress_db      Created      0.0s
WARNING: The "wordpress" service is using the custom container name "wordpress_app"
. Docker requires each container to have a unique name. Remove the custom name to scale the service
```

Result: 3 WordPress containers running.

Problem: All try to use same port (8080), no load balancing.

Solution: Use Reverse Proxy (Nginx). Add another service, then configure load balancing manually.

```

yuvraj_singh@LUCIF3R: ~/
GNU nano 7.2 docker-compose.yml *
version: '3.9'

services:
  nginx:
    image: nginx:latest
    ports:
      - "8080:8080"

  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

```

Limitations of Compose Scaling:

- * No built-in load balancing
- * No auto-healing
- * Single host only
- * Not production-ready for scaling

6. Running Same Setup with Docker Swarm

Step 1: Initialize Swarm: `docker swarm init`

```

yuvraj_chawa@LUCiFSR:~/expo/Wp-compose-lab? docker swarm init advertise-addr
2.22.87.136
Swarm initialized: current node (ify2vksfmucLt85lars33nqa5) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-165kee8g9o2a50elsje4o1uh6gmpnan3ddg5mfikvzt5
    tenzus-35q154kjdtalvxnor01vs8ij8 172.22.87.136:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow th
e instructions

```

Step 2: Deploy Stack: `docker stack deploy -c docker-compose.yml wpstack`

```
yuvraj_singh@LUCIF3R: exp6 wp-compose-labf docker stack deploy ~C docker-compose
.yml wpstack
Ignoring unsupported options: restart

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
Creating service wpstack_nginx
```

Step 3: Scale Service: `docker service scale wpstack_wordpress=3`

```
anirudh_chawla@LUCIF3R: ~/exp6/wp-compose-lab$ docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service wpstack_wordpress converged
```

What Changes in Swarm?

Feature	Docker Compose	Docker Swarm
Scope	Single host	Multi-node cluster
Scaling	Manual	Built-in
Load balancing	No	Yes (internal LB)
Self-healing	No	Yes
Rolling updates	No	Yes
Networking	Basic	Overlay network

7. Benefits of Docker Swarm

- * Built-in load balancing
- * Automatic container restart (self-healing)
- * Horizontal scaling across nodes
- * Rolling updates without downtime
- * Service abstraction (not individual containers)

8. Challenges / Limitations of Swarm

- * Less popular than Kubernetes
- * Limited ecosystem
- * Less flexible scheduling
- * Fewer enterprise features

9. Key Learning Outcomes

- * Multi-container apps require orchestration
- * Docker Compose is ideal for: Development, Testing, Learning
- * Docker Swarm is useful for: Simple production clusters, Easy scaling without Kubernetes complexity

10. Conclusion

This experiment demonstrated:

- * How to deploy WordPress + MySQL using Docker Compose
- * How containers communicate using internal networking
- * Importance of volumes for persistence
- * Scaling limitations of Compose
- * Advantages of using Docker Swarm for production-ready deployments

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